Deep Transformer based Structural Semantic Network for Document Clustering

***Abstract*—Document clustering is very sensitive to document representation schemes. Traditional document clustering approaches used document content to model features and used representation based on these features thus suffer from problems like vocabulary mismatch and word sense ambiguities, and fail to cater with word to word relationship in longer dependency. Consequently, clustering results lack semantic values. Deep document clustering employs deep neural networks to simultaneously learn semantic representation and clustering. In this paper, we propose a deep transformer based structural semantic network for document clustering (DTSN), it uses a transformer based approach to learn a document-level semantic representation with long term text dependency, and exploits graph convolution network in order to capture semantic between documents in a layer by layer fashion utilizing reinforcement learning. Extensive series of experiments are performed and our proposed approach performs substantially better than state-of-the-art deep document clustering approaches. We report a new state-of-the art on unsupervised text document clustering tasks.**

I. Introduction

Document Clustering is an essential technique in the world of natural language processing which aims to organize vast amounts of documents in distinct groups based on their underlying and inherent similarities. It helps with summarizing, efficient information retrieval and understanding of large textual datasets among other things. Traditional approaches to document clustering are quite lackluster as they fail to grasp the similarities between documents.

With the recent boom in information on the internet and the rise of availability in datasets, document clustering has seen a rise in research and interest as it can identify the underlying features and structures of these documents. In recent years, deep document clustering has seen quite some interest as it overcomes some of the difficulties faced by traditional approaches.

One such idea was to map data samples to a lower dimension hidden space so as to extract important features and find data partitions. Majority of these approaches also only focus on the internal representations of documents extracted from the semantics learned. However, this proves problematic as documents are limited in their length as well as express their meaning in various ways. This approach leads to the models extracting some of the meaningful features but also missing some of the important ones.

We have also observed that the shortcomings of the majority of these models is the inability to grasp long term dependency between words which leads to an inferior understanding of the underlying structure of the document. Alongside this, important information can also be discovered from the outside of the document. Close neighbours of documents that are regarded as the external structural information can also provide vital information that can help in deciding data partitions. Therefore, it can be beneficial for us to use this information to enhance our internal semantic representation.

In this paper, we design a complete end-to-end document clustering model namely DTSN, that learns document partition using an ensemble structurally enhanced network. Our approach enhances the internal representation of a document using the external representation gathered from its neighbours to calculate an ensemble enhanced representation which is then used to calculate document partitions.

For each document, an Encoder only Transformer is used to calculate the internal semantic representation through the usage of Multi-Head Attention, Positional Encoding and Position-Wise Feed-Forwarding Networks. The common strategy is the usage of Autoencoders(AE) [12] to map the document to a lower dimension however we found that the usage of Transformers and their ability to grasp long term dependency produced better results. For the external representation of each document, we employed the use of Graph Convolution Networks(GCN) [13] as they showed good results for the displaying of documents as graphs and the encoding of each data sample with the relevant external information. Both these models learn different but important semantic information that aids in document clustering. The favoured strategy to enhance the internal representation with the external one is for both approaches to run independently and then combine in the end using a normal fusion process with the assumption that the two processes are independent of each other [14]–[16], however, the information that both processes contain can help aid each other in producing a more robust representation. As such, we decided on a layer by layer enhanced ensemble approach that learns the internal representation as well as the enhanced ensemble representation in every layer by combining the internal and external representations.

The enhanced representation is then used to further reinforce the internal representation of the given document. Furthermore, we also used an objective function to not only calculate data partitions but also to optimize the document representation learning networks of the DTSN model.

The 2 main contributions of this paper are as follows:

- An end-to-end transformer based deep document clustering model namely DTSN, performs document clustering using a structural enhanced network. It specifically uses internal semantics extracted from a transformer encoder and enhances them using external semantics extracted from a GCN.
- Extensive experiments on real world datasets are carried out against other state-of-the-art clustering datasets to show the effectiveness of the DTSN model. The results to these experiments indicate the substantial improvement DTSN makes in comparison to the other models.

II. Related Works

This paper is related to and takes inspiration from 3 branches of related work, in particular, Transformers, Deep Clustering using internal semantics and Deep Clustering using external semantics. We will also discuss some traditional approaches to document clustering.

Over the years, Document Clustering has faced many problems, dealing with sparse amount of text in documents, documents expressing the same meaning in different approaches, underlying semantics of the documents, relations between different documents and long term dependencies found in them. Traditionally, numerous approaches have been tried. [41] compares the basic k-means algorithm with bisecting k-means and agglomerative clustering to figure out which is better on numerous datasets. [39] uses Lexical Chains to group related words together alongside WordNet to extract semantics to overcome to limitations of bag-of-words models and capture deeper semantic relationships. [42] leverages frequent word sequences for effective document clustering, it also implements cluster candidate formations and cluster merging. [43] uses NLP techniques like dependency parsing and named-entity recognition to locate propositions which it uses as the fundamental units of information for clustering.

In recent years, the transformer's self-attention mechanism, that allows the model to give relative importance to certain parts of the data, has increasingly become the focal point of various research studies. [3] investigates the ability of the self-attention mechanism of BERT in classifying scientific articles and assesses the subset of the most attended words with other feature selection methods in order to identify self-attention as a viable feature selection approach. BERT architecture is further employed by [5] to convert radio-logical reports into embedding vectors to classify these reports by priority. [4] utilizes transformer architecture in natural language processing to predict chemical compound retro-synthesis using text-like embedding of chemical reactions. [6] introduces probabilistic transformers that can transform an arbitrary data embedding into uni-variate Gaussian Mixtures while ensuring low metric distortion. Furthermore, [7] presents Contrastive Learning with Bi-directional transformer (CBiT) which enhances sequential recommendation by utilizing bidirectional transformers to process long user sequences sliding window approach creating fine grained division of user sequences.

Likewise, the usage of deep neural networks in clustering has seen a rise as they perform well when it comes to learning feature representations and providing clustering assignments. The Deep clustering network(DCN) [11] uses dimensional reduction techniques using a deep neural network as well as k-means for clustering. Deep embedding clustering(DEC) [8] maps the data space to a lower dimension embedding space which is optimized using KL-divergence-based objective, this was improved by the improved deep embedding clustering(IDEC) [9] which added local structures. Variational deep embedding(VDE) [10] can learn the document representation from the document generation process which can then be used for document clustering. Similarly, there are many other models [17]–[22] that have proposed models using internal semantics, however they haven't tackled the issue of sparsity of text or the insufficient representations created for document clustering by only learning internal features. None of them have also tried to leverage external representations.

To grasp at the underlying semantics in the external representations, the popular approach has been the use of spectral clustering [23]. Which creates a graph with samples as nodes and uses that graph structure as the basis for clustering. Deep neural network for learning graph representations(DNGR) [24] uses spectral clustering with adjacency matrices for similarity matrices. In recent years however, the task of graph clustering has progressed to finding related sets of vertices in graphs. In these approaches, samples are represented by nearby graph structures modeled by the deep neural network [24], [25]. The GCN has also seen great success in this task as many approaches have used it for graph clustering [26]–[33]. The graph autoencoder(GAE) and the variational graph autoencoder(VGAE) [26] have used 2-layer GCNs to learn the semantic representation and then use the autoencoder and the variational autoencoder to reconstruct the adjacency matrix. These two approaches were further improved by the adversarially regularized graph autoencoder(ARGA) and the adversarially reg-

ularized variational graph autoencoder(ARVGA) [27] respectively by incorporating an adversarial method into the two approaches. All of these graph based approaches however fail to utilize the internal semantic information of documents for the creation of data partitions.

There have been however, some models, that consider both the external and internal information for reinforcing the clustering process. [32], [33] The structural deep clustering network(SDCN) [32] uses a GCN and an AE and uses a delivery system to reinforce the GCN with the internal representation learned by the AE. Based on this, the attention-driven graph clustering(AGCN) [33] created a system to fuse the two representations by the aggregation of multi-scale features embedded at different layers. The deep structural enhanced network for document clustering(DSEDC) [40] further improved on the SDCN and introduced the creation of an enhanced ensemble representation by combining the external and internal representation, layer by layer and using the enhanced representation to further improve the internal representation at later layers. SDCN and AGCN however fall short due to their focus on the GCN-based structural representation, as this contains a large number of noise for text documents and the content semantics that are learned are largely ignored. The DSEDC model employs the usage of AEs which do a great job at learning internal semantics but largely fail at grasping long term dependencies, something that transformers are designed to do well.

III. Our Proposed model: DTSN

In this section, We provide details for our proposed DTSN model by explaining the overall framework followed by individual modules and optimizations.

A. Overall Framework

The framework for our proposed model consists of two modules, the GCN Module and the Transformer Module. For an input document data sample,The GCN module creates a reasonable data partition using its closeness. There are multiple layers of the GCN model which outputs an external data representation.

The Transformer module is used to calculate the enhanced internal document representation and the ensemble representation while keeping in mind both the internal and external representations. In the first layer, it uses the internal representation it learned with the external representation learned by the GCN to create the ensemble document representation. The internal document representation is further improved through reinforcement learning by using the ensemble representation as the input for the next layer to learn the internal representation so in the I-th layer, the internal representation is used to create the ensemble representation which is used as input for the internal representation in the I+1-th layer.

The ensemble representation is used as input for the Document Clustering module alongside the external representation from the GCN module, this module uses them to create document partitions as well as using a joint objective function to further fine tune the results alongside the representations.

B. Transformer Encoder Module

Transformer network architecture is a novel way to tackle sequence to sequence modeling tasks. Its self-attention mechanism excels at capturing dependencies within the data regardless of whether those connections happen over time (temporal) or space (spatial); allowing significantly more parallelization than its counterparts. In this study, the encoder layer of the transformer network is investigated to capture and produce context aware internal structural embeddings for each document sample.

In order for the transformer's encoder layer to make use of order of sequence, information about the relative and absolute positions of the input data is injected into the input sequence. We have used the sine and cosine functions to generate the positional encodings where pos is the position and i is the dimension:

$$\mathrm{PE}(pos, 2i) = \sin\left(\frac{\mathrm{pos}}{1000^{2i/d_{\mathrm{model}}}}\right)$$

$$\mathrm{PE}(pos, 2i+1) = \cos\left(\frac{\mathrm{pos}}{1000^{2i/d_{\mathrm{model}}}}\right)$$

Positional encoding is added to the input sequence before sending it to the encoder stack.

A transformer encoder stack can consist of N identical layers where each layer consists of 2 sublayers. First layer is the multi-head self-attention mechanism, and the second layer is position wise fully connected feed-forward network. Residual connections are employed around each sublayer, followed by a normalizing layer.

The multi-attention layer receives the input and projects it down into key, query, and value vectors. Multi-head attention enables the model to jointly compute information from different representations of subspaces at different positions. Key, value, and query vectors are linearly projected h times with different, learned linear projections; on which each of them the attention function is performed in parallel.

An attention function maps each query vector to a set of key-value pairs to an output vector. The output is computed as a weighted sum of values vector, where weights corresponding to each value is calculated by a compatibility function of the query with the corresponding key vector. Attention function on the set of queries

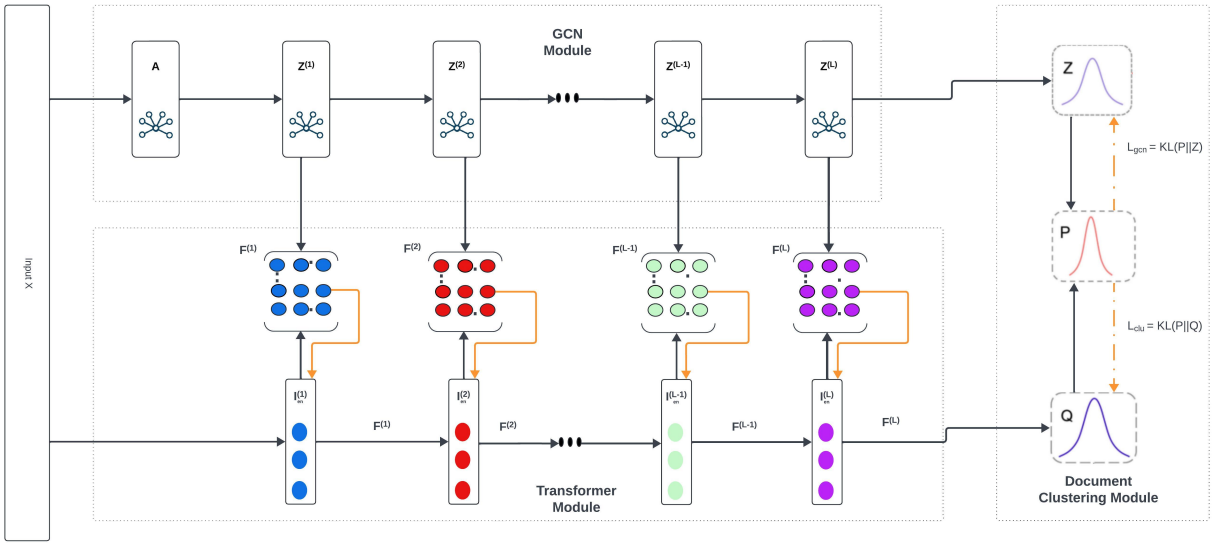

Fig. 1. The DTSN model

is computed simultaneously along with key, values, and dimension of key vector dk as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{\text{dk}}}\right)V$$

Final representation is the concatenation of each output from the attention function. The output from the multi-head attention layer is then normalized before sending it to the position-wise feed forward network where a fully connected feed-forward network is applied on each position separately and identically but having different parameters from layer to layer. This contains 2 linear transformations with a ReLU activation in between. The result of this layer is further normalized before it is passed through a series of linear transformations to map its latent space that may result in enhanced representation of context rich internal embeddings.

C. GCN Module

In the GCN module, Graph Convolutional Networks (GCNs) have a crucial role in capturing external structural representations of each document sample. Each document representation Z(l) at l-th layer is calculated through the following convolution calculation:

$$Z^{(l)} = f(Z^{(l-1)}, A|W_c^{(l-1)})$$
$$f(Z^{(l-1)}, A|W_c^{(l-1)}) = \phi_c(\overline{D}^{-\frac{1}{2}}\overline{AD}^{-\frac{1}{2}}Z^{(l)}W_c^{(l)})$$
$$\overline{A} = \overline{A} + I$$
$$\overline{D}_{ii} = \sum_{j=1}^{N}\overline{A}_{ij}$$

where the GCN structure of each data sample is represented by A. D_{ii} is the degree matrix of of the GCN structure and W_c^l is the weight matrix at the GCN's l-th layer . Various strategies have been utilized to assess the similarity for each document sample, both with or without explicit graph structure. For samples having clear graph structures, GCNs have been made using the respective graph structure. For the remaining samples, not having defined graph representation, KNN model has been used to determine K-nearest neighbors and their similarities are leveraged to construct the GCN structure.

D. DC Module

The DC module is subdivided into two tasks. First is to learn the document partitions using the enhanced document representation generated with the help of the transformer module. The second task is to optimize the document representation capturing networks of our proposed model. An objective function is being employed which makes use of confidence of clustering partitioning to achieve both tasks all together. For learning document partition, Student's-distribution has been used as a kernel to measure similarity between the enhanced ensemble document representations and the cluster centers. The clustering distribution Q and the targeting distribution P are being calculated as follows:

$$q_{ij} = \frac{(1 + \|f_i - \mu_j\|^2/\alpha)^{\frac{-\alpha+1}{2}}}{\sum_{j'=1}^{J}(1 + \|f_i - \mu_{j'}\|^2/\alpha)^{\frac{-\alpha+1}{2}}}$$

$$p_{ij} = \frac{q_{ij}^2/\tau_j}{\sum_{j'=1}^{J} q_{ij}^2/\tau_j}$$

where τ_j is the soft cluster frequency, α is the degree of freedom of the Student's distribution, f_i is the ensemble document representation and μ_j is the cluster center. Using these distributions the objective function of our document clustering model is derived as follows:

$$L_{clu} = KL(P\|Q) = \sum_{i=1}^{N}\sum_{j=1}^{J} p_{ij} log\frac{p_{ij}}{q_{ij}}$$

where N and J represent the number of documents and number of clusters respectively.

We also use the same method for optimization of the GCN module by calculating similarity between external representation z_i and μ_j, the distribution is as follows:

$$z_i j = \frac{(1 + \|z_i - \mu_j\|^2/\alpha)^{\frac{-\alpha+1}{2}}}{\sum_{j'=1}^{J}(1 + \|z_i - \mu_{j'}\|^2/\alpha)^{\frac{-\alpha+1}{2}}}$$

where Z is the clustering result distribution created by the Z^L layer. Similarly, we use the targeted distribution P to guide the GCN with the following objective function.

$$L_{gcn} = KL(P\|Z) = \sum_{i=1}^{N}\sum_{j=1}^{J} p_{ij} log\frac{p_{ij}}{z_{ij}}$$

With the final loss being as follows:

$$L = L_{clu} + w_g L_{gcn}$$

Optimization is achieved through tuning the document representational networks. The objective function combines clustering loss, and GCN tuning loss to improve document clustering accuracy and representational learning. We also use the Ranger optimizer [34] that combines the RAdam optimizer [37] and the LookAhead optimizer [38] to optimize cluster centers and network parameters.

IV. Experiments

A. Datasets

We will conduct our experiments on two real-world datasets, the descriptions are below:

- ACM[1]: From the ACM website, it is a document dataset that includes links. Features are bag of words of the keywords and list of coauthor links between documents. Papers published are divided into 3 classes(database, wireless communication and data mining)
- BBC[2]: BBC is a web text document dataset with 5 classes of news, business, entertainment, politics, sports and tech. It consists of 2225 documents from the BBC website containing news stories from 2004-2005

B. Baseline Methods

- Methods using internal representation only:
  K-means [35] is a traditional clustering model without deep learning.
  Lexical Chains [39] are made using semantics from WordNet and are used as features for document clustering
  AE [12] and VAE [36] use autoencoders and variational autoencoders respectively to learn representations for clustering
  DEC [8] is a deep clustering model that designs a clustering objective to guide the learning of the data representations.
  IDEC [9] improves DEC by adding a reconstruction loss.
- Methods using external representation only:
  Spectral [23] treats the samples as the nodes in a weighted graph and uses the graph structure of data for clustering.
  DNGR [24] is a spectral clustering method that uses the adjacency matrix as the similarity matrix.
- Methods using both internal representation and external representation:
  GAE and VGAE [26] are unsupervised graph embedding models using GCN to learn representations on the autoencoder and variational autoencoder respectively.
  ARGA and ARVGA [27] improve GAE and VGAE by adding an adversarial model.
  SDCN [32] is a structural deep clustering model that combines a GCN and a DNN to to construct a clustering network
  SDCNQ [32] is the variant of SDCN with distribution Q.
  AGCN [33] created a system to fuse the two representations by the aggregation of multi-scale features embedded at different layers,
  DSEDC [40] is a enhanced clustering network that combines a GCN and AE to create an enhanced representation to construct a clustering network
  DSEDC-S3 [40] is a variation of the DSEDC model where the 3 layers are enhanced from the beginning layer

C. Experimental Setup

For the Baseline methods with pretraining processes(AE, DEC, IDEC, SDCN, AGCN, DSEDC), the same dimensions are used: i-500-500-200-10 where i is

[1]http://dl.acm.org/
[2]http://mlg.ucd.ie/datasets/bbc.html

the input data dimension. They're trained for 30 epochs with 10^{-3} learning rate by the Adam algorithm. For all baseline methods, we use the results reported in the given papers. For our model, the dimensions of the GCN and Transformer are all equal to the input data's dimensions with 5 layers, with K = 30 on non-graph datasets as well as α = 2, γ = 0.5 and w_g = 0.01. For the ACM dataset, Positional Encoding in the transformer is used while in the bbc dataset it is not. We train the DTSN model for 300 epochs with 10^{-4} learning rate using the Ranger algorithm.

D. Results

This section shows the results of our model on the given datasets in comparison to the baselines. Table 1 shows the results on two datasets, one with links and one without.

As can be observed, in all three metrics, our DTSN model outperforms the other baseline models by a wide margin, these results provide substantial evidence to advocate for our proposed DTSN model. Discussing the results, our model improves on the DSEDC model by 11.59%, 27.2% and 30.12% on the BBC dataset with respect to Accuracy, NMI and ARI. It also improves on the DSEDC-S3 model by 7.88%, 32.84% and 25.62% on the ACM dataset with respect to Accuracy, NMI and ARI.

TABLE I
CLUSTERING RESULTS ON REAL-WORLD DOCUMENT DATASETS

Methods	BBC			ACM		
	ACC	NMI	ARI	ACC	NMI	ARI
Kmeans	51.15	32.97	14.37	67.31	32.44	30.60
AE	51.73	51.33	37.33	81.83	49.30	54.64
VAE	70.23	53.04	43.04	69.75	31.16	32.53
DEC	77.06	61.51	59.59	84.33	54.54	60.64
IDEC	76.40	64.48	61.72	85.12	56.61	62.16
Spectral	71.96	61.68	43.73	73.62	40.63	40.89
DNGR	76.90	56.21	50.16	37.69	1.76	1.63
GAE	79.75	65.54	61.69	84.52	55.38	59.46
VGAE	79.91	63.43	63.23	84.13	53.20	57.72
ARGA	84.31	61.56	65.16	83.50	51.09	56.04
ARVGA	85.44	64.73	66.75	85.45	54.86	61.22
SDCN_Q	77.02	62.97	60.91	86.95	58.90	65.25
SDCN	77.55	65.28	62.58	90.45	68.31	73.91
AGCN	72.60	62.98	57.38	90.59	68.38	74.20
DSEDC	88.61	75.35	74.78	91.33	70.99	76.14
DSEDC-S3	88.16	72.25	71.25	92.32	73.72	78.66
DTSN	**98.88**	**95.85**	**97.31**	**99.60**	**97.93**	**98.81**

TABLE II
ABLATION STUDY

Model Varaints	ACM		
	ACC	NMI	ARI
Tr	75.50	68.14	57.90
Gr	35.01	0.13	0.01
Tr+Gr	86.58	69.79	66.63
Tr+Gr+Er	99.60	97.93	98.81

E. Ablation Study

An Ablation Study is conducted on the ACM dataset to validate that each component of our model contributes to its effectiveness. For this study, Tr will refer to the transformer module, Gr refers to the GCN module and Er refers to the ensemble learning per layer. For the information in Table 2, Tr utilizes only the transformer module, Gr only utilizes the GCN module, Tr+Gr utilize both but ensemble representation is only created at the end with no layer by layer reinforcement and Tr+Gr+Er is our complete DTSN module. As can be seen in the table, one module on its own cannot provide the results given by our full model. Tr lacks the external representation provided by the Gr module and the layer by layer reinforcement. Gr lacks the internal representation gained by the Tr module while Tr+Gr cannot provide good results as the layer by layer reinforcement is absent. This proves each part of the model clearly contributes to its performance and the full model, Tr+Gr+Er clearly outperforms all baselines further cementing the effectiveness of our approach.

F. Visualization of Clusters during Training Process

To explore how to clusters evolve over the training process, we analyze the clusters formed in all datasets in this section. In Figure 2 and 3, we can see, the clusters are not quite clear and the data points are quite connected to each other but over time as we can see, clear clusters are formed. For ACM, we can see clear separation between the 3 clusters at the end of the training process while the clusters formed in the BBC dataset are not quite as clear. However, there is clear enough seperation to form distinct clusters and provide good results in both datasets.

V. CONCLUSION

In this paper, a state-of-the-art end-to-end deep document clustering model namely DTSN, was presented to further improve on the document semantic data representation problem through an ensemble reinforcement based

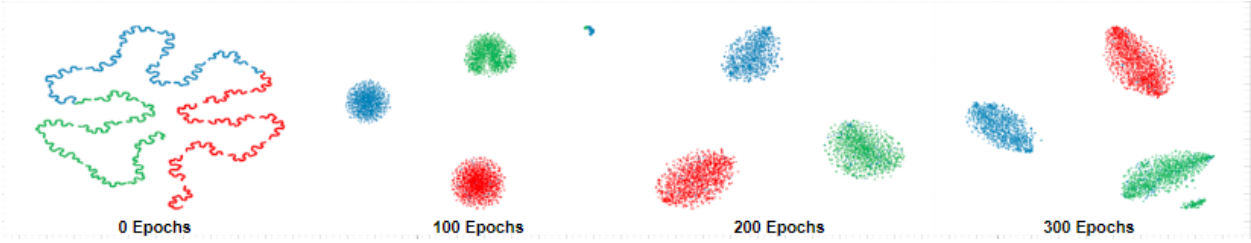

Fig. 2. ACM Clusters

Fig. 3. BBC Clusters

strategy to capture both internal and external representations and the use of transformers to improve the internal representation. A joint objective function was also used to simultaneously improve the clustering and learned representations. Experiments conducted on multiple real world datasets clearly show the effectiveness of our model.

For further work, this model can be adjusted and worked on further to implement on other real-world datasets. Further improvements can also be made to the NMI and ARI score for those datasets as the model can be lacking in that regard.

VI. Bibliography

References

[1] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[2] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.

[3] Garcia-Silva, A., Gomez-Perez, J.M. (2021). "Classifying Scientific Publications with BERT - Is Self-attention a Feature Selection Method?". In: Hiemstra, D., Moens, MF., Mothe, J., Perego, R., Potthast, M., Sebastiani, F. (eds) Advances in Information Retrieval. ECIR 2021. Lecture Notes in Computer Science(), vol 12656. Springer, Cham. https://doi.org/10.1007/978-3-030-72113-8_11

[4] Tetko, I.V., Karpov, P., Van Deursen, R. Goden, G. "State-of-the-art augmented NLP transformer models for direct and single-step retrosynthesis". Nat Commun 11, 5575 (2020). https://doi.org/10.1038/s41467-020-19266-y

[5] S. Ayanouz, A. Boudhir, M. Ahmed, "Unleashing the Power of NLP and Transformers: A Game-Changer in Medical Research and Clinical Practice and a revolution of Medical Text Analysis.," May 2023, doi: https://doi.org/10.1145/3607720.3607787.

[6] A. Kratsios, V. Debarnot, and I. Dokmanić, "Small transformers compute universal metric embeddings," Mar. 06, 2024. https://dl.acm.org/doi/10.5555/3648699.3648869

[7] H. Du, H. Shi, P. Zhao, D. Wang, V. Sheng, Y. Liu, G. Liu, L. Zhao,"Contrastive Learning with Bidirectional Transformers for Sequential Recommendation," Proceedings of the 31st ACM International Conference on Information & Knowledge Management, Oct. 2022, doi: https://doi.org/10.1145/3511808.3557266.

[8] Xie J, Girshick R, Farhadi A (2016) Unsupervised deep embedding for clustering analysis. In: International conference on machine learning, p PMLR

[9] Guo X, Gao L, Liu X, Yin J (2017) Improved deep embedded clustering with local structure preservation. In: Ijcai, pp 1753–1759

[10] Jiang Z, Zheng Y, Tan H, Tang B, Zhou H (2017) Variational deep embedding: an unsupervised and generative approach to clustering. In: IJCAI

[11] Yang B, Fu X, Sidiropoulos ND, Hong M (2017) Towards k-means-friendly spaces: simultaneous deep learning and clustering.
In: International conference on machine learning. PMLR, pp 3861–3870

[12] Hinton G, Salakhutdinov RR (2006) Reducing the dimensionality of data with neural networks. Science 313(5786):504–507

[13] Kipf TN, Welling M (2017) Semi-supervised classification with graph convolutional networks. In: 5th International conference on learning representations

[14] Bai R, Huang R, Chen Y, Qin Y (2021) Deep multi-view document clustering with enhanced semantic embedding. Inform Sci 564:273–287

[15] Shi S, Nie F, Wang R, Li X (2020) Auto-weighted multi-view clustering via spectral embedding. Neurocomputing 399:369–379

[16] Li Y, Liao H (2021) Multi-view clustering via adversarial view embedding and adaptive view fusion. Appl Intell 51(3):1201–1212

[17] Li Y, Liao H (2021) Multi-view clustering via adversarial view embedding and adaptive view fusion. Appl Intell 51(3):1201–1212

[18] McConville R, Santos-Rodriguez R, Piechocki RJ, Craddock I (2021) N2d:(not too) deep clustering via clustering the local manifold of an autoencoded embedding. In: 2020 25th International conference on pattern recognition (ICPR). IEEE, pp 5145–5152

[19] Ji Q, Sun Y, Hu Y, Yin B (2021) Variational deep embedding clustering by augmented mutual information maximization. In: 2020 25th International conference on pattern recognition (ICPR). IEEE, pp 2196–2202
[20] Xia W, Zhang X, Gao Q, Gao X (2021) Adversarial self-supervised clustering with cluster-specificity distribution. Neurocomputing 449:38–47
[21] Wang R, Li L, Wang P, Tao X, Liu P (2021) Feature- aware unsupervised learning with joint variational attention and automatic clustering. In: 2020 25th International conference on pattern recognition (ICPR). IEEE, pp 923–930
[22] Zhang B, Qian J (2021) Autoencoder-based unsupervised clustering and hashing. Appl Intell 51(1):493–505
[23] Ng AY, Jordan MI, Weiss Y (2002) On spectral clustering: analysis and an algorithm. In: Advances in neural information processing systems, pp 849–856
[24] Cao S, Lu W, Xu Q (2016) Deep neural networks for learning graph representations. In: Proceedings of the AAAI conference on artificial intelligence, vol 30
[25] Hu P, Chan KC, He T (2017) Deep graph clustering in social network. In: Proceedings of the 26th international conference on world wide web companion, pp 1425–1426
[26] Kipf TN, Welling M (2016) Variational graph auto-encoders. Statistics 1050:21
[27] Pan S, Hu R, Fung S-f, Long G, Jiang J, Zhang C (2019) Learning graph embedding with adversarial training methods. IEEE Trans Cybern 50(6):2475–2487
[28] Qin J, Zeng X, Wu S, Tang E (2021) E-gcn: graph convolution with estimated labels. Appl Intell 51(7):5007–5015
[29] Ou G, Yu G, Domeniconi C, Lu X, Zhang X (2020) Multi-label zero-shot learning with graph convolutional networks. Neural Netw 132:333–341
[30] Kou S, Xia W, Zhang X, Gao Q, Gao X (2021) Self-supervised graph convolutional clustering by preserving latent distribution. Neurocomputing 437:218–226
[31] Xia W, Wang Q, Gao Q, Zhang X, Gao X (2021) Self-supervised graph convolutional network for multi-view clustering. IEEE Transactions on Multimedia
[32] Bo D, Wang X, Shi C, Zhu M, Lu E, Cui P (2020) Structural deep clustering network. In: Proceedings of the web conference 2020, pp 1400–1410
[33] Peng Z, Liu H, Jia Y, Hou J (2021) Attention-driven graph clustering network. In: Proceedings of the 29th ACM international conference on multimedia, pp 935–943
[34] Wright L (2019) Ranger - a synergistic optimizer. In: GitHub Repository, https://githuvb.com/lessw2020/Ranger-Deep-Learning- Optimizer. Accessed 12 June 2021
[35] Hartigan JA, Wong MA (1979) Algorithm as 136: a k-means clustering algorithm. J R Stat Soc Series c (Appl Stat) 28(1):100–108
[36] Kingma DP, Welling M (2014) Auto-encoding variational bayes. Statistics 1050:1
[37] Liu L, Jiang H, He P, Chen W, Liu X, Gao J, Han J (2019) On the variance of the adaptive learning rate and beyond. In: International conference on learning representations
[38] Zhang M, Lucas J, Ba J, Hinton G (2019) Lookahead optimizer: k steps forward, 1 step back. Adv Neural Inf Process Syst, 32 Multi-scale graph attention subspace clustering network. Neurocomputing 459:302–314
[39] Tingting Wei, Yonghe Lu, Huiyou Chang, Qiang Zhou, and Xianyu Bao. 2015. A semantic approach for text clustering using WordNet and lexical chains. Expert Syst. Appl. 42, 4 (March 2015), 2264–2275. https://doi.org/10.1016/j.eswa.2014.10.023
[40] Lina Ren, Yongbin Qin, Yanping Chen, Ruina Bai, Jingjing Xue, and Ruizhang Huang. 2022. Deep structural enhanced network for document clustering. Applied Intelligence 53, 10 (May 2023), 12163–12178. https://doi.org/10.1007/s10489-022-04112-z
[41] M. Gupta and A. Rajavat, "Comparison of Algorithms for Document Clustering," 2014 International Conference on Computational Intelligence and Communication Networks, Bhopal, India, 2014, pp. 541-545, doi: 10.1109/CICN.2014.123.
[42] Li, Yanjun Chung, Soon Holt, John. (2008). Text Document Clustering Based on Frequent Word Meaning Sequences. Data Knowledge Engineering. 64. 381-404. 10.1016/j.datak.2007.08.001.
[43] Ernst, O., Caciularu, A., Shapira, O., Pasunuru, R., Bansal, M., Goldberger, J., Dagan, I. (2022). Proposition-Level Clustering for Multi-Document Summarization. arXiv preprint arXiv:2112.08770.